

Predicted Packet Padding for Anonymous Web Browsing Against Traffic Analysis Attacks

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Abstract—Anonymous communication has become a hot research topic in order to meet the increasing demand for web privacy protection. However, there are few such systems which can provide high level anonymity for web browsing. The reason is the current dominant dummy packet padding method for anonymization against traffic analysis attacks. This method inherits huge delay and bandwidth waste, which inhibits its use for web browsing. In this paper, we propose a predicted packet padding strategy to replace the dummy packet padding method for anonymous web browsing systems. The proposed strategy mitigates delay and bandwidth waste significantly on average. We formulated the traffic analysis attack and defense problem, and defined a metric, cost coefficient of anonymization (CCA), to measure the performance of anonymization. We thoroughly analyzed the problem with the characteristics of web browsing and concluded that the proposed strategy is better than the current dummy packet padding strategy in theory. We have conducted extensive experiments on two real world data sets, and the results confirmed the advantage of the proposed method.

Index Terms—Anonymity, predicted packet padding, web browsing.

I. INTRODUCTION

THE purpose of this paper is to present a novel strategy, predicted packet padding, to serve anonymous web browsing systems in order to make them usable in practice by mitigating delay and bandwidth waste. Anonymous communication has been explored in the last two decades with the focus on protecting user privacy in the Internet age. According to a recent survey [1], anonymous communication systems can often be classified into two categories: high-latency systems and low-latency systems. High-latency anonymity systems are able to pro-

vide strong anonymity, but are typically only applicable for non-interactive applications that can tolerate delays of several hours or more, such as the mix networks [2] for e-mail messages. On the other hand, low-latency anonymity systems often provide better performance and are intended for real-time applications, particularly for web browsing (in this paper we use low-latency anonymity systems and anonymity web browsing systems interchangeably).

There are a few implementation mechanisms for current low-latency anonymous systems, such as the onion routing [3], [4], the Tor system [5], [6] and the Crowds system [7]. Most of them employ the dummy packet padding strategy against traffic analysis attacks. There are a number of metrics proposed to measure anonymity of a given system, such as the information theory-based metrics [8], [9].

The current anonymous web browsing systems are far from perfect and are vulnerable against traffic analysis attacks. The purpose of attacks on anonymous communication is to identify pairwise entities in a system rather than the content of communication. For example, does user A communicate with user B? Does user C access website D? In general, every web page is distinct from others in terms of measurable features, such as the length of its HTML text, number of web objects, number of packets for each web object, and the timing information of packet transportation. These features form a *fingerprint* from which each unique web page can be identified. Furthermore, the fingerprint of a website is also unique because it is a combination of all the fingerprints of its web pages.

Encryption on web pages is vulnerable against traffic analysis attacks in terms of anonymity. A website may be accessed in an encrypted way, e.g., using SSL, but traffic encryption is vulnerable to traffic analysis. Sun *et al.* [10] found that encryption does not hide all information about the encrypted web pages with HTTP object count and their sizes often revealing rich information about the encrypted web pages. Liberatore and Levine [11] used a similarity method to infer the source of encrypted HTTP streams, and they achieved up to 90% accuracy in real world tests. Zhu *et al.* [12] employed a flow correlation-based traffic analysis to attack mix based low-latency anonymous networks, and found that all but a few batching strategies failed against their attacks. There have been a number of recent reports of successful attacks on anonymous networks [13]–[16].

The time constraint is a critical challenge for anonymous web browsing. The dummy packet padding strategy is widely used by defenders against traffic analysis attacks, however this method introduces huge delays and bandwidth waste. In order to hide the fingerprint of a protected web page or website, dummy packets are usually injected into the intended traffic to disguise

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the unique fingerprint of the web page or web site [17], [18]. According to Shannon's perfect secrecy theory [19], we have to inject a large number of dummy packets in order to achieve a high level of anonymity, which in turn causes huge delay and bandwidth waste [20]. Various strategies were proposed to improve the performance of anonymous systems, such as the dependent link padding algorithm [21] and the game theory-based algorithm [22].

In this paper, instead of using dummy packets to achieve anonymization, we propose a different padding strategy, predicted packet padding to do so. The proposed method disguises the fingerprints of web pages at the server side by injecting predicted web pages, which users are going to download, as cover traffic. Our proposal comes from the following fact: for one browsing session, web viewers usually access multiple web pages at one web site. Huberman *et al.* [23] found that the average browsing length for a browsing session was 15 pages at the end of the 1990s. Xie and Yu's recent investigation demonstrated that the average browsing length for a session is around 30 [24]. Moreover, this fact has been confirmed by the applications of web caching and web page prefetching technologies [25], [26]. With this fact in place, we can therefore use predicted web pages to replace dummy packets for packet padding. This strategy can significantly mitigate the problem of extra delay and extra bandwidth demand of dummy packet padding.

The contributions of this paper are summarized as follows.

- 1) We propose the predicted packet padding method to replace the current dummy packet padding one for implementation of anonymous web browsing systems. This new strategy can significantly reduce the average delay caused by dummy packet padding, and therefore, makes anonymous web browsing applicable in practice.
- 2) Applying the proposed method for a given web site, we can prevent an eavesdropper from identifying the pages that a web browser has downloaded through traffic analysis. The proposed method can also be extended to protect other objects, such as web sites.
- 3) We explore the problem of anonymization efficiency for the first time to the best of our knowledge. Previous works have only presented various metrics for anonymity measurement. In this paper, we define a metric, the cost coefficient of anonymization (CCA), to measure the efficiency of different padding strategies.
- 4) We establish a mathematical model for the traffic analysis attack and defense problem for anonymous web browsing systems. This model paves a foundation for further analysis and comparison amongst different anonymization strategies for the research community.

The rest of the paper is structured as follows. The related work is discussed in Section II. We present the setting of the attack and defense problem in Section III, followed by modeling the system in Section IV. We analyze the performance of the proposed method and the dummy packet padding method in Section V. The implementation details of the predicted packet padding strategy are presented in Section VI. Experiments based on real world data sets are conducted in Section VII. We present a further discussion in Section VIII. Finally, we summarize the paper and present future work in Section IX.

II. RELATED WORK

A. Anonymous Web Browsing

The HTTP protocol [27] shows that when a client submits an HTTP request of an URL, the corresponding server will deliver the HTML text to the client, and the HTML text includes the references of the related objects, e.g., images or flashes. The objects will be downloaded to the client one after the other. This information composes fingerprints of web pages. Moreover, an observer can clearly see packet heads, which include critical information of the traffic, such as source IP addresses and destination IP addresses.

The measurement of information anonymity has been explored for decades. Shannon [19] investigated the conditions for perfect secrecy and used entropy as a metric to measure anonymity. He concluded that a system is in the best anonymity when the system entropy achieves in maximum, namely every object in the system is the same. However, Fu *et al.* [28] pointed out that constant rate traffic padding cannot achieve perfect secrecy as it cannot satisfy the conditions for Shannon's perfect secrecy due to its implementation mechanism. Diaz, *et al.* [8] introduced the degree of anonymity, a definition based on entropy, to quantify the anonymity of connections. Serjantov and Danezis [9] also proposed an information theoretical measure of anonymity that takes into account the probabilities of users sending and receiving messages.

Researchers have demonstrated that data encryption is not effective against traffic analysis attacks. Sun *et al.* [10] tried to identify encrypted network traffic using the HTTP object number and size, and their experiments indicated that these two features are sufficient to identify a significant fraction of the world wide websites. Following this direction, Wright *et al.* [29] confirmed that websites can be identified with high probability even if it is an encrypted channel. Coull *et al.* [30] evaluated the strength of the anonymization methodology in terms of preventing the assembly of behavioral profiles and concluded that anonymization offers less privacy to web browsing traffic than what people expected. Liberatore and Levine [11] used a profiling method to infer the sources of encrypted HTTP connections. They used packet length and direction as attributes and established a profile database for individual encrypted network traffic. The match technique integrated a similarity metric (Jaccard's coefficient) and a supervised learning technique (the naive Bayesian classifier). They were able to infer the source of individual encrypted network traffic based on their method. Their extensive experiments showed that the proposed method can identify the source with an accuracy of up to 90%.

A few solutions have been proposed for anonymous web browsing against traffic analysis attacks. Hintz [17] suggested to inject noise traffic (also called *cover traffic* in some papers) to the intended traffic which changes the fingerprints of the actual traffic, and transparent pictures were employed to add extra fake connections against fingerprint attacks. Wright *et al.* [31] recently proposed a traffic morphing method to protect the anonymity of communication. They transformed the intended website's (e.g., www.webmd.com) fingerprint to the fingerprint of another website (e.g., www.espn.com). Their transformation methods included packet padding and packet splitting. Optimal

techniques were employed to find the best cover website (in terms of minimum cost for transformation) from a list. They tested their algorithm against the data set used in Liberatore and Levine's work [11] and found that the proposed method can improve the anonymity of website access and reduce the overhead at the same time. All of this research is based on the strategy of dummy packet padding.

Time constraint is a critical challenge for the current anonymous systems. Wang *et al.* [21] proposed the minimizing sending rate for a given set of flows by controlling the generation rate of the covering traffic; therefore, the traffic pattern of the communication channel is the same all the time. As a result, full anonymity is obtained according to Shannon's perfect secrecy theory. Venkitasubramaniam [18] noticed the delay caused by adding dummy packets into communication channels and proposed transmission schedules on relay nodes to maximize network throughput to obtain a desired level of anonymity. Venkitasubramaniam and Tong [22] recently introduced game theory into the optimization of communication anonymization in wireless environments. The problem is modeled as a two player zero-sum game between the network designer and the adversary. Once again, all these efforts are under the framework of dummy packet padding strategy.

B. Web Browsing Characteristics

Web caching and prefetching are effective and efficient solutions for web browsing when bandwidth is a constraint. Awad *et al.* tried to predict a user web surfing path using a hybrid model, which combined the Markov model and support vector machine learning. The fusion of the two models complements each other's weakness and worked well. Montgomery *et al.* [32] found that the information of a user's web browsing path offers important information for a transaction, and a successful deal is usually followed by a number of the same path being accessed. Teng *et al.* [25] argued that there must be elaborate coordination between client side caching and prefetching and formulated a normalized profit function to evaluate the profit from caching an object. The proposed function integrated a number of factors, such as object size, fetching cost, reference rate, invalidation cost and invalidation frequency. Their event-driven simulations demonstrated that the proposed method performed well.

Breslau *et al.* [33] analyzed web accessing behavior and found that it followed a Zipf-like distribution. Suppose there are n web pages in total, and they are sorted by popularity from the most to the least as $w_1, w_2, \dots, w_n$. Let $p(w_i)$ be the access probability of the i th page, then

$$p(w_i) = \frac{\Omega}{i^{\alpha_z}} \quad (1)$$

where α_z is the Zipf index, which varies from trace to trace. When $\alpha_z = 1$, (1) becomes the Zipf distribution.

A more general form of the distribution is called the Zipf-Mandelbrot distribution [34], which is defined as follows:

$$p(w_i) = \frac{\Omega}{(i + q)^{\alpha_z}} \quad (2)$$

where the additional constant q ($q \geq 0$) is called the plateau factor, which makes the probability of the highest ranked objects flat. Since $\sum_{i=1}^n p(w_i) = 1$, $\Omega = (\sum_{i=1}^n (1)/((i + q)^{\alpha_z}))^{-1}$.

The Zipf-Mandelbrot distribution becomes the Zipf distribution when $q = 0$. Hefeeda and Saleh [35] found that the Zipf-Mandelbrot distribution is more suitable for caching of peer-to-peer systems.

Klemm *et al.* [36] explored caching in finite and infinite request stream cases, respectively. If there have been r web pages in the cache, then the hit-ratio $h(r)$ for the next request for web page i is as follows:

$$h(r) = \sum_{i=1}^n p(w_i) [1 - (1 - p(w_i))^r] \quad (3)$$

where n is the total number of web pages.

It is widely accepted that the size distribution of web object obeys the Pareto distribution [37], which is defined as follows. Let X be a random variable and x_m be the minimum size of web objects. For a given web object, the probability of its size being greater than x is

$$p(X > x) = \left(\frac{x_m}{x}\right)^{\alpha_p} \quad (4)$$

where α_p is a positive parameter and called the Pareto index.

Some researchers have indicated that the distribution of the size of web object is composed of two distributions: a log-normal distribution for the body up to a transition point, and a Pareto distribution for the tail [37], [38]. Reed *et al.* [39] revised this into one formula, referring to as double Pareto distribution. This finding was supported by Burklen and his colleagues [40]. However, the difference between the double Pareto distribution and the Pareto distribution is at the tail part, with the majority of the two distributions similar to each other.

Moreover, in terms of mathematics, both the Pareto distribution and the Zipf distribution belong to the power law. For the same objects of power law, we can use any one of them to represent it, and the difference is that there is a -1 difference between the Pareto index and the Zipf index. We refer interested readers to [41].

III. PROBLEM STATEMENT

We set our problem as follows: An adversary (Bob) knows that the monitored user (Alice) accesses a web site, but wants to know which pages she has browsed. In order to protect Alice's privacy, the web site has to take two actions. The first action is to encrypt the intended pages to hide the fingerprints, and secondly, packet padding should be used to disguise the fingerprints of the intended pages. As a user, Bob has the full knowledge of all the web pages of the web site in the forms of encrypted or plain text. We suppose Bob captures all network traffic on Alice's computer. Bob expects to use traffic analysis methods to identify the web pages that Alice has assessed.

We can extend this problem to a higher scale: Bob wants to identify which web sites Alice has accessed from a list of possible web sites. For example, Alice surfs the web through an anonymous system, and Bob expects to identify the web sites

that she has accessed through traffic analysis. These two problems are essentially the same with different computation scales, and therefore we only discuss the former in this paper.

For a given web site, we assume that there are n ($n > 0$) web pages, denoted as $\{w_1, w_2, \dots, w_n\}$, where w_i is the i th ($1 \leq i \leq n$) popular web page. For a given web page w_i ($1 \leq i \leq n$) of the web site, we suppose that it possesses m ($m > 0$) web objects, $\{w_i^1, w_i^2, \dots, w_i^m\}$. In general, we denote w_i^k ($1 \leq i \leq n, 1 \leq k \leq m$) as the k th web object of the i th web page; and we denote the size (in terms of packets) of web object w_i^k ($1 \leq i \leq n, 1 \leq k \leq m$) as $|w_i^k|$.

Furthermore, we denote the fingerprint of a web page as a set, $\{t_i^1, t_i^2, \dots, t_i^m\}$. Each element of the set, t_i^k ($1 \leq i \leq n, 1 \leq k \leq m$), is defined as follows:

$$t_i^k = \frac{|w_i^k|}{\sum_{j=1}^m |w_i^j|}. \quad (5)$$

The *a priori* of Alice's accessing of w_i ($1 \leq i \leq n$) is denoted as $p(w_i)$ ($1 \leq i \leq n$), which represents the probability that web page w_i is accessed by Alice among all the n web pages, and it is natural that the following holds:

$$\sum_{i=1}^n p(w_i) = 1. \quad (6)$$

Bob knows $p(w_i)$ ($1 \leq i \leq n$) along with the complete information of the fingerprint of every web page of the web site. Alice browses a number of web pages through encrypted channels, therefore, Bob cannot read the content of the packets. However, Bob is able to monitor Alice's local network and can observe her ongoing browsing in terms of the number of packets for each web object

$$\tau = \{\tau_1, \tau_2, \tau_3, \dots\}$$

where τ_i is the number of packets of the encrypted i th web object that Alice has downloaded.

In general, every web page is different from others in terms of its page fingerprint. For example, each web page possesses a different number of web objects, and each web object may have a different size. This fingerprint information is reachable to Bob, and encryption will not change the fingerprint information too much as indicated by the previous work [10]–[12]. Based on his observation and the Bayesian theorem, Bob is able to claim that Alice accessed web page w_i with the following probability:

$$p(w_i|\tau) = \frac{p(w_i) \cdot p(\tau|w_i)}{p(\tau)}$$

where $p(w_i)$, $p(\tau|w_i)$ and $p(\tau)$ are known to Bob as he can use the same method to access the same web site to obtain the information.

As we know from previous work, data encryption cannot hide the fingerprints of web pages. Therefore, we have to introduce other methods, such as packet padding, to decrease $p(w_i|\tau)$ as much as we can. According to Shannon's perfect secrecy theory [19], an adversary cannot break the anonymity if the following equation holds:

$$p(w_i|\tau) = p(w_i). \quad (7)$$

This means the observation offers no information to the adversary. In terms of implementation, the web server needs to make every web page the same to Bob, meaning from Bob's view, every traffic session has the same number of web objects and every web object has the same size. The current method to achieve this is dummy packet padding, which causes huge delay and bandwidth waste. This inherent disadvantage incumbers the usage of anonymous systems in interactive applications, e.g., web browsing.

IV. PREDICTED PACKET PADDING MODELING

Based on the previous discussion, we know that web page data encryption by itself cannot defeat traffic analysis attacks, and packet padding can achieve anonymity for web browsing even though it is not practical because there are huge delays and extreme bandwidth cost caused by the dummy packet padding mechanism. We therefore combine data encryption and predicted packet padding, rather than dummy packet padding, at the server side to achieve feasible anonymous web browsing.

In the practice of traffic analysis, eavesdroppers employ various features to break anonymity, such as packet arrival time interval, number of packets of different web objects. In this paper, we aim to present the novelty and effectiveness of the proposed predicted packet padding strategy and therefore confine our discussion within the following reasonable conditions.

- 1) We only discuss the traffic analysis attack on the fingerprint of web object size (number of packets). We must note that the proposed method is also effective against other types of attacks that use other kinds of page fingerprint, such as packet arrival time interval.
- 2) We suppose cache is in place at client computers. The memory of major computing devices, such as PC, notebook, are sufficient to offer caching. Moreover, the major web browsers, e.g., IE and Firefox, use cache browsing as the default setting.

The proposed predicted packet padding mechanism is shown in Fig. 1. At the server side, the predicted web pages Y will be used as cover traffic for packet padding for the intended page W . The output of the anonymization, Z , will be encrypted and transported to the client via the Internet or related anonymous networks. At the client side, once Z is decrypted, the intended page W will be displayed by a web browser, and the predicted pages Y will be deposited at the local cache. We use symbol $\oplus$ to represent the packet padding operation.

The details are depicted as follows: As a client, Alice submits an encrypted HTTP request for web page w_i ($1 \leq i \leq n$) to the web server. The web server will return the intended traffic $w_i = \{w_i^1, w_i^2, \dots, w_i^m\}$, where w_i^k ($1 \leq k \leq m$) represents the k th web object of web page w_i , and the size of the object w_i^k is $|w_i^k|$ in terms of packets.

In general, in order to disguise the fingerprint of page w_i , we need to make every web object look the same to eavesdroppers. A common method is to inject cover traffic $y_i = \{y_i^1, y_i^2, \dots, y_i^{m'}\}$, where y_i^k ($1 \leq k \leq m', m \leq m'$) represents the cover traffic for w_i^k ($1 \leq k \leq m$) into the intended traffic w_i to obtain a covered output $z_i = \{z_i^1, z_i^2, \dots, z_i^{m'}\}$.

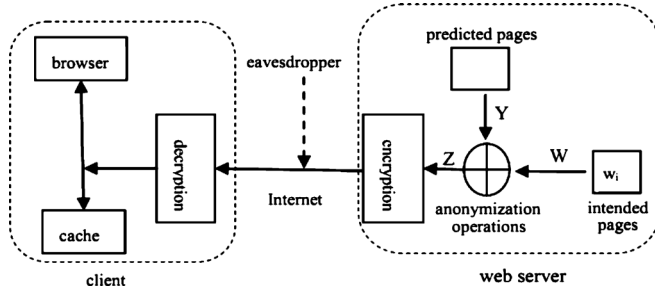


Fig. 1. Anonymous web browsing system model with packet padding using predicted pages.

TABLE I
SYMBOLS IN THIS PAPER

Notation	Description
W	Random variable for intended traffic
Y	Random variable for cover traffic
Z	Random variable for covered traffic
w_i	The web page i
$ w_i $	The size of w_i (in terms of packets)
w_i^k	The k^{th} web object in web page w_i
$ w_i^k $	The size of w_i^k (in terms of packets)
y_i^k	The cover traffic for w_i^k
$ y_i^k $	The size of y_i^k (in terms of packets)
z_i^k	The covered traffic for w_i^k
$ z_i^k $	The size of z_i^k (in terms of packets)
N	Average of number of object for a page
S	Average of size of web objects
$C(Y)$	Cost for traffic Y
$C(Y W)$	Cost for Y in order to cover traffic W
$\oplus$	Packet padding operation

As a result, the observation, $\tau = \{\tau_1, \tau_2, \tau_3, \dots\}$, which Bob obtains, is the encrypted version of $\{z_i^1, z_i^2, \dots, z_i^m\}$.

The notation of the cover traffic and the covered traffic is similar to the notations of the intended traffic. For the sake of convenience, we summarize the symbols that we use in this paper in Table I. Following Table I, the anonymization operation at the server side is as follows:

$$Z = W \oplus Y. \quad (8)$$

If we further breakdown (8) for an intended traffic w_i , then the following equation holds:

$$|z_i^k| = |w_i^k| + |y_i^k|, k = 1, 2, 3, \dots, m'. \quad (9)$$

In order to defeat Bob, we need to make every browsing session the same from Bob's viewpoint. This means every session has the same number of web objects and every web object has the same size, namely, the following two equations hold:

$$\begin{cases} |z_i| = |z_j| & i, j = 1, 2, 3, \dots, i \neq j \\ |z_i^{k_1}| = |z_i^{k_2}| & k_1, k_2 = 1, 2, 3, \dots, k_1 \neq k_2. \end{cases} \quad (10)$$

Different from the traditional dummy packet padding strategy, we use the predicted web pages that users are going to download in the near future as the cover traffic. At Alice's site, after the decryption, the intended data W goes to the web browser, and the cover traffic Y goes to the cache.

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Following the web browser cache mechanism, Alice's following requests will be checked with the cache first, if the requested web page is in the cache, then there is no need to download it from the server again, and there is no request for the page from Alice to the server. As a result, the goal of anonymity in web browsing is served and the delay issue is addressed from a long term perspective.

It is true that we cannot predict the expected web pages 100% accurately for clients, therefore, part of the prefetched pages may never be requested, and such pages cause bandwidth waste and delay as dummy packets do. In order to measure the efficiency of anonymization operation, we define a metric as follows.

Cost Coefficient of Anonymization (CCA): Let function $C(X)$ represent the cost for network traffic X . For an intended network traffic W , we inject a cover traffic Y to achieve anonymity, then the cost coefficient of the anonymization operation is defined as

$$\beta = \frac{C(Y|W) + C(W)}{C(W)}$$

where $C(Y|W)$ denotes the cost of traffic Y , which is used to cover an intended traffic W .

In this paper, we define the cost function $C(X)$ as the number of packets for a given network traffic X . Suppose a user has browsed k ($k = 1, 2, \dots$) pages, then the cost coefficient of anonymization is defined as

$$\beta = \frac{\sum_{i=1}^k (|y_i| + |w_i|)}{\sum_{i=1}^k |w_i|} = \frac{\sum_{i=1}^k |z_i|}{\sum_{i=1}^k |w_i|}. \quad (11)$$

We have to note that CCA indicates the efficiency of packet padding. In other words, what percentage of the prefetched web pages is actually browsed by the user in terms of the average. At the same time, CCA also demonstrates the extra delay because of the nonbrowsed web pages that we prefetched.

The essential problem now is how to compose the covered traffic Z with cover traffic Y , given the intended traffic W . We expect to make Y as small as possible in order to make the current downloading time as short as possible under the condition of anonymity.

We can transfer the problem into the following optimization problem:

$$\begin{aligned} & \min. \sum_{k=1}^l |y_i^k| \\ & \text{s.t.} \\ & |z_i| = |z_j|, \quad i, j = 1, 2, 3, \dots, i \neq j \quad (a) \\ & |z_i^{k_a}| = |z_j^{k_b}|, \quad i, j, a, b = 1, 2, 3, \dots \quad (b). \end{aligned} \quad (12)$$

In the optimization equation (12), condition (a) implies that every browsing session has the same size, and condition (b) guarantees that every covered web object in the same or different browsing sessions possesses the same size.

In order to achieve traffic anonymity as we discussed in the previous section, we have to make every response looks the same to an adversary: the same number of web objects and every web object has the same size. Based on previous research, we

know the size of web objects follows the Pareto distribution, namely a few super large ones followed by many small size web objects. It is very expensive if we use the size of the largest object and the number of objects of the most crowded page as a benchmark. Our previous investigation [20] indicated that the cost will be on the 10^2 order.

In this paper, we forge a standard page and use this standard page as the benchmark for packet padding. We assume that the standard page has $\bar{N}$ objects and the size of each object is $\bar{S}$, where $\bar{N}$ is the statistical average of the number of web objects on a page for a given web site, and $\bar{S}$ is the statistical average of web object size for the web site. They are defined as follows:

$$\bar{N} = \sum_{i=1}^n \left(p(w_i) \cdot \sum_{j=1}^m I_{s_j} \right) \quad (13)$$

where n is the total number of web pages, m is the number of objects of a given page, and function I_{s_j} is defined as follows:

$$I_{s_j} = \begin{cases} 1, & s_j \in w_i \\ 0, & s_j \notin w_i \end{cases}$$

where s_j is the j th object of a web page, $\sum_{j=1}^m I_{s_j}$ represents the number of web objects of a page

$$\bar{S} = \sum_{i=1}^n \left(p(w_i) \cdot \sum_{j=1}^m |s_j| \right) \quad (14)$$

where $|s_j|$ represents the size of the web object s_j .

In an ideal situation, the intended page possesses $\bar{N}$ objects and each object's size is $\bar{S}$, and the same for the cover traffic. Therefore, each session of downloading possesses $2\bar{N}$ web objects and each web object's size is $\bar{S}$ packages. As a result, the standard size of one data downloading session is as follows:

$$|Z| = |W| + |Y| = 2\bar{N} \cdot \bar{S}. \quad (15)$$

V. ANALYSIS ON PREDICTED PACKET PADDING METHOD

In this section, we are going to analyze the proposed method following Section IV. We first investigate the effectiveness of the proposed method. In practice, people use various devices to surf the web. Some browsers use desktops or notebooks, which possess sufficient cache size, usually at the Gigabyte level. Moreover, the cache space demanded for one browsing session is usually limited as the browsing length is limited. Therefore, the cache size is "infinite" in this case. On the other hand, more and more web users use wireless mobile devices to surf the web, such as the iPhone, iPad, etc. These devices have small memory for caching, we therefore treat these cases as finite cache size. We will analyze these two cases respectively in this section.

A. Effectiveness of Proposed Method

As we discussed in the previous sections, the designated method is proposed to fight against web page fingerprint based

attacks. Suppose a protected web site has n web pages in total. In the case that an attacker knows nothing about the page popularity, when we apply the proposed anonymization method, every page downloaded from the web site is the same in terms of page fingerprint. Therefore, the probability of a successful inference of the downloaded page is $1/n$. However, the page popularity is usually known to attackers, but it is hard for them to know our padding strategy.

We define *anonymization effectiveness (AE)* as a metric of the probability of a successful anonymization as follows:

$$AE(m) = 1 - p_a(m) \quad (16)$$

where m is the number of web pages a browser has accessed, and $p_a()$ represents the probability of successful inference of an attacker.

Based on our padding strategy, when Alice has browsed m pages from the protected web site, the eavesdropper will roughly observe $k = \lceil (1/(2))m \rceil$ downloading sessions. They will naturally believe that Alice has browsed k pages as they do not know that we are using the predicted packet padding strategy.

We therefore obtain the anonymization effectiveness of the proposed method as follows:

$$AE_{\text{predict}}(m) = 1 - \sum_{j=1}^{\lceil \frac{1}{2}m \rceil} p(w_j) \quad (17)$$

where w_j is the j th popular web page.

At the same time, the anonymization effectiveness of the dummy packet padding strategy is

$$AE_{\text{dummy}}(m) = 1 - \sum_{j=1}^m p(w_j). \quad (18)$$

Comparing (17) and (18), we find that

$$AE_{\text{predict}}(m) = AE_{\text{dummy}}(m) + \sum_{j=1+\lceil \frac{1}{2}m \rceil}^m p(w_j). \quad (19)$$

It is easy to see that the predicted packet padding strategy is always better than the traditional dummy packet padding as $\sum_{j=1+\lceil (1)/(2)m \rceil}^m p(w_j)$ is usually greater than zero. A numerical comparison between these two padding methods in terms of anonymization effectiveness is shown in Fig. 2.

Intuitively, the total number of web pages has a big impact on AE of the proposed method. For example, if the protected web site has a small number of web pages, then the advantage of the proposed method is limited compared to the dummy packet padding. We further present numerical examples in Fig. 3 to demonstrate this. The results show that our anonymization method is more effective for web sites with a large number of web pages. Moreover, the fewer pages a user browses, the better the protection is.

B. Infinite Cache Size

In this section, we analyze the proposed method in the case of client computers with infinite cache size for web browsing.

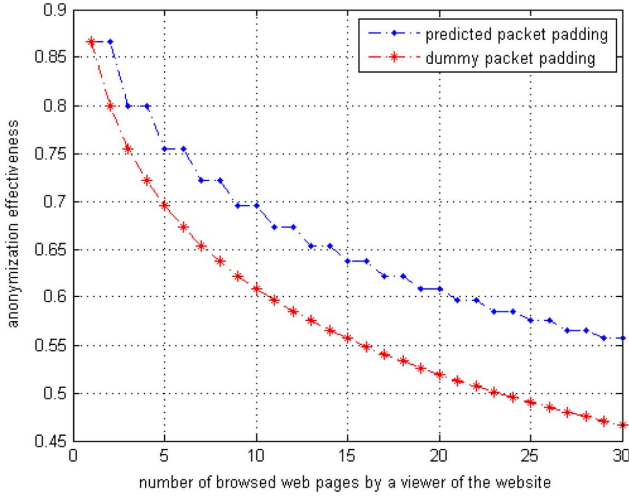


Fig. 2. Comparison of anonymization effectiveness between dummy packet padding and proposed predicted packet padding, where $\alpha_z = 1.0$ and total number of web pages is 1000.

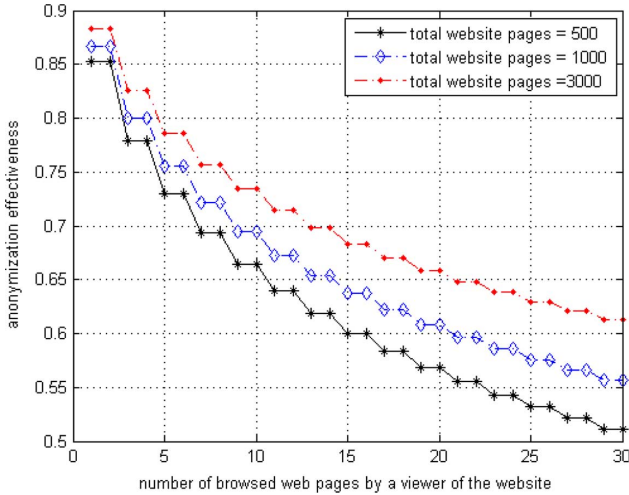


Fig. 3. Anonymization effectiveness of proposed method against number of browsed pages under different total number of web pages as 1000, 5000, and 10000, respectively, and $\alpha_z = 1.0$.

Suppose a client has browsed r web pages of a given web site, and the volume of the downloaded traffic is $|w_{i_1}|, |w_{i_2}|, \dots, |w_{i_r}|$, which is denoted by $|W_r|$ as follows:

$$|W_r| = \sum_{x=1}^r |w_{i_x}|.$$

When r is sufficiently large, e.g., $r > 10$, we know that the probability of the r intended pages are the r most popular web pages is $\sum_{i=1}^r p(w_i)$, namely,

$$\sum_{i=1}^r p(w_i) = \frac{\sum_{i=1}^r |w_i|}{|W_r|}. \quad (20)$$

Under the same circumstances, the volume of the cover traffic can be expressed as follows:

$$\begin{aligned} |Y_r| &= \sum_{x=1}^r |y_{i_x}| \\ &= (1 - h(r)) |W_r| \\ &\leq \left(1 - \sum_{i=1}^r p(w_i)\right) \cdot |W_r| \end{aligned} \quad (21)$$

where $h(r)$ is the hit ratio, which is defined in (3).

By combining (20) and (21) with the definition of CCA in (11), we can estimate the CCA when a user has browsed r pages

$$\begin{aligned} \beta(r) &= 1 + \frac{(1 - h(r)) \cdot |W_r|}{\sum_{i=1}^r p(w_i) \cdot |W_r|} \\ &\approx 1 + \frac{1 - \sum_{i=1}^r p(w_i)}{\sum_{i=1}^r p(w_i)} \\ &= \frac{1}{\sum_{i=1}^r p(w_i)}. \end{aligned} \quad (22)$$

From (22), we obtain the following two properties for $\beta(r)$:

$$\begin{cases} \beta(r_1) \leq \beta(r_2), & r_1 > r_2 \\ \lim_{r \rightarrow \infty} \beta(r) = 1. \end{cases}$$

The first property shows that the cost coefficient of anonymity decreases when we browse more pages in a browsing session. The second property indicates that the cost coefficient of anonymity is close to one when the browsing length is sufficiently large.

We can also obtain the expression of CCA for the dummy packet padding strategy in the infinite cache size case. We also suppose that a user has browsed r pages, and combined with (15), the size of the covered traffic is

$$|Z'_r| = r \cdot 2 \cdot \bar{N} \cdot \bar{S}. \quad (23)$$

When r is sufficient, we can use the most popular r pages to replace the actual intended traffic, namely,

$$|W'_r| \approx \sum_{i=1}^r p(w_i) \cdot |w_i|. \quad (24)$$

Combining (23), (24) and the definition of CCA in (11), when r is sufficiently large, e.g., $r > 10$, we obtain the estimation of CCA for the dummy packet padding β_d as follows:

$$\beta_d(r) = \frac{|Z'_r|}{|W'_r|} \approx \frac{2r \cdot \bar{N} \cdot \bar{S}}{\sum_{i=1}^r p(w_i) \cdot |w_i|} \approx 2. \quad (25)$$

It is natural that we are interested in when the proposed strategy can beat the dummy packet padding one in terms of browsing length, namely, $\beta(r) < \beta_d(r)$. Combining (22) and

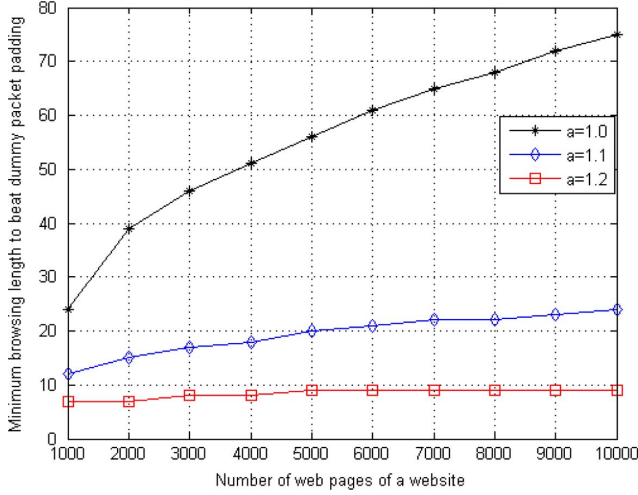


Fig. 4. Minimum browsing length of predict packet padding strategy to beat dummy packet padding method with various Zipf factor α and total number of web pages of a web site.

(25), we obtain the minimum browsing length, l , which is determined by the following condition:

$$\sum_{i=1}^l p(w_i) > 0.5. \quad (26)$$

We know $p(w_i)$ follows the Zipf-Mandelbrot distribution, and it mainly depends on two parameters: the skewness factor α and the total number of pages n . In Fig. 4, we show the minimum browsing length for the proposed strategy to beat the traditional dummy packet padding method in theory. We can see that the skewness factor α has an obvious impact on the result. Our method outperforms the dummy packet padding even when the browsing length is less than 10 for a 9000 page web site when $\alpha = 1.2$.

C. Finite Cache Size

In the case that clients have finite cache space, especially a small cache size (e.g., mobile devices), we must arrange the predicted pages well, otherwise, the prefetched pages will be replaced by the later prefetched pages before they are referred, which will cause the same result as the dummy packet padding strategy.

A simple and effective way for this case is to cache the most popular pages within the limited cache space. Suppose a client can cache c web pages ($c \approx (\text{cache size})/(\bar{N} \cdot \bar{S})$, which we refer to as the *relative cache size*). As a result, the hit ratio is

$$h(c) = \sum_{i=1}^c p(w_i) = \sum_{i=1}^c \frac{\Omega}{(i+q)^{\alpha_z}}. \quad (27)$$

When a user has browsed r web pages, and r is sufficiently large, combined with the definition in (11), we can calculate CCA for finite cache sizes in two cases: When $c \leq r$, the CCA

is the same as (22). In the case that $c > r$, we can derive the CCA as follows:

$$\begin{aligned} \beta(r) &= 1 + \frac{|Y_r|}{|W_r|} \\ &= 1 + \frac{1 - h(c)}{\sum_{i=1}^r \Pr(w_i)} \\ &= 1 + \frac{\sum_{i=1}^r p(w_i) - \sum_{i=1}^c p(w_i)}{\sum_{i=1}^r p(w_i)} \\ &= 2 - \frac{\sum_{i=1}^c p(w_i)}{\sum_{i=1}^r p(w_i)}. \end{aligned} \quad (28)$$

Combining (22) and (28), we obtain the CCA for finite cache size c as

$$\beta(r) = \begin{cases} \frac{1}{\sum_{i=1}^r p(w_i)}, & r \leq c \\ 2 - \frac{\sum_{i=1}^c p(w_i)}{\sum_{i=1}^r p(w_i)}, & r > c \end{cases}. \quad (29)$$

Following (29), we have $\beta(r) \approx \beta_d(r) \approx 2$ when $c \rightarrow 0$, namely, the CCA of predicted packet padding for very small cache size is similar to the CCA of dummy packet padding.

For the dummy packet padding strategy, the $\beta_d(r)$ for this case is the same for the infinite cache size case, which has been expressed in (25).

VI. PREDICTED PACKET PADDING ALGORITHM

We will present the algorithms for the proposed anonymization method in this section. We first proposed a simple but effective algorithm to decide which pages will be used for padding. Following this, we present the predicted packet padding algorithm.

A. Page Prediction Algorithm

We need an effective and simple algorithm to identify the cover pages for padding, which are usually the pages that a web browser is going to view.

There are some advanced web page prediction algorithms in the literature [25], [26], however, the cost of the current prediction algorithms is expensive. In order to improve their prediction accuracy, a system must maintain the global statistics and browsing history of every client. In our case, we propose a simple but effective prediction algorithm for packet padding. Based on the page popularity, from a system viewpoint, we know that the most popular pages are of the highest probability in being requested sooner or later. Therefore, we can ignore the relationship between the latest requested page and the next requested page, which will reduce the computing workload significantly, while at the same time maintaining the performance of prediction in our packet padding case.

The basic idea of our prediction algorithm is as follows. We establish a queue of predicted pages, which includes the most popular m (m is sufficiently large) pages sorted according to their popularity. If a requested page is not one of the m pages, take the first page in the predicted queue as the cover page for the requested page; If the requested page is one of the m pages,

Algorithm 1: Web Page Prediction Algorithm

```

//establish the predicted queue;
1. Initialize a predicted queue  $Q_p$  with length  $m$  ;
2. Populate  $Q_p$  with the most popular  $m$  pages, sorted by
   popularity, and  $head = 1$  ;
while true do
  3. A page  $W_i$  is requested ;
  //if the requested page is in the queue, take it off;
  4. if  $W_i \in Q_p$  then
    |  $Q_p = Q_p - \{W_i\}$  ;
  end
  //the predicted page  $Y_i$  for intended page  $W_i$ ;
  5.  $Y_i = Q_p(head)$  ;
  6.  $head = head + 1$ ;
end

```

Algorithm 2: Predicted Packet Padding Algorithm

```

//calculating the standard download size;
1.  $|Z_s| = 2\bar{N} \cdot \bar{S}$ ;
while true do
  for A new user  $i$  do
    while user  $i$  request page  $j(1 \leq j \leq n)$  do
      2.  $|Y_j| = |Z_s| - |w_j|$ ;
      //identifying padding pages;
      3. call algorithm 1 until  $|Y_j|$  is met;
      //padding with predicted pages ;
      4. using the packets of  $|Y_j|$  to meet
         requirement of equation (10);
    end
  end
end

```

then take the requested page off from the predicted queue, and take the first page in the queue as the cover page.

The algorithm is shown in Algorithm 1.

B. Predicted Packet Padding Algorithm

In this section, we present the detail of the packet padding algorithm with the predicted pages obtained from Algorithm 1. The task of this algorithm is to make the intended traffic all the same using the predicted pages. As we analyzed in the previous section, we can use the same predicted packet padding algorithm for both infinite and finite cache size cases. For the finite cache size case, the only difference is that when the cache is full, the result of our algorithm is the same as the dummy packet padding method. We therefore only present the algorithm for the case of infinite cache size in this section. The algorithm is shown in Algorithm 2.

VII. PERFORMANCE EVALUATION

In order to evaluate the advantages of the proposed strategy, we studied two popular web sites of China. For privacy protection reasons, we denote them as web site U and A. We collected the page requests of the two web sites, number of web objects for each active page and the related size of all web objects for one week at a major network center, which covers about a quarter of the total network traffic of China. In detail, we studied 9081 active pages of web site U, and 7765 active pages of web site A. The characteristics that we were interested are presented as follows.

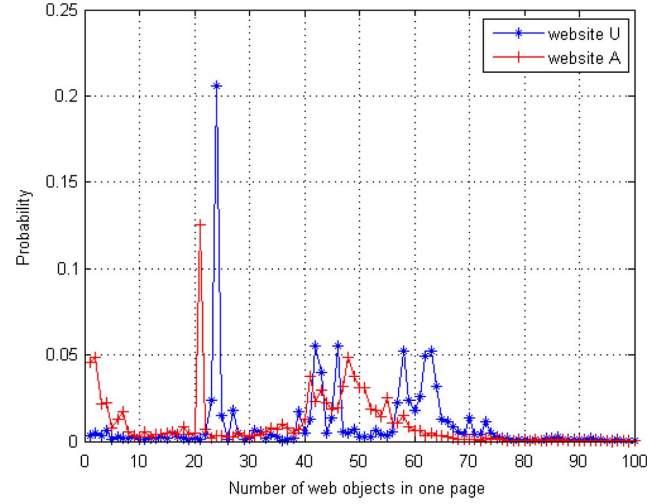


Fig. 5. Probability distribution of number of web objects in one page for two popular Chinese web sites.

First, we were interested in gaining an idea of how many objects are usually contained in a web page, namely the distribution of the number of objects. We counted the number of web objects in each page of the two data sets and unified the distribution. The results are shown in Fig. 5.

Based on Fig. 5, we found that the probability distribution of the number of web objects in web pages is quite uneven, and there are four outstanding segments for each data set. A page with around 25 objects has the highest probability, followed by 40–50 and 55–65 objects in a page respectively, with both of the data sets having an obvious number of pages with less than 10 objects in a page.

Secondly, we wanted to know the web object size distribution of the two data sets. We measured the size of every web object in the two data sets. Based on previous studies, we know that it follows the Pareto distribution. In the two data sets, we collected 11 198 different web objects from web site U, and 6651 from web site A. We ranked the objects according to their size and present them in a loglog diagram in Fig. 6. We do see the Pareto distribution for web object size for both web sites.

Thirdly, we expected to know the web page popularity of the two web sites. We observed the two web sites for one week and recorded web page requests. The results are presented in Fig. 7.

We observed the Zipf-Mandelbrot distribution for both web sites in Fig. 7. Moreover, in order to obtain the parameters α_z and q for the two web sites, we also did a simulation to extract the parameters. We found that the best matched parameters for the two web sites are $\alpha_z = 1.2$ and $q = 8$. These parameters will be used in the following experiments.

Figs. 6 and 7 look similar; the reason is that both the Pareto distribution and the Zipf-like distribution are special cases of the power law distribution.

For the purpose of comparison, we took the same number of web pages from the two data sets for further experiments. We know that the highest ranked pages dominate the major percentage of the studied domain, and our two data sets have 9081 and 7765 active web pages, respectively. In order to find out the accessing probability coverage over the web pages, we calcu-

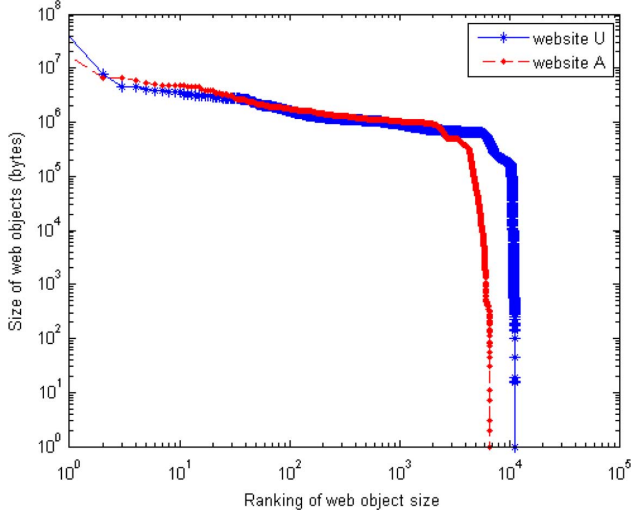
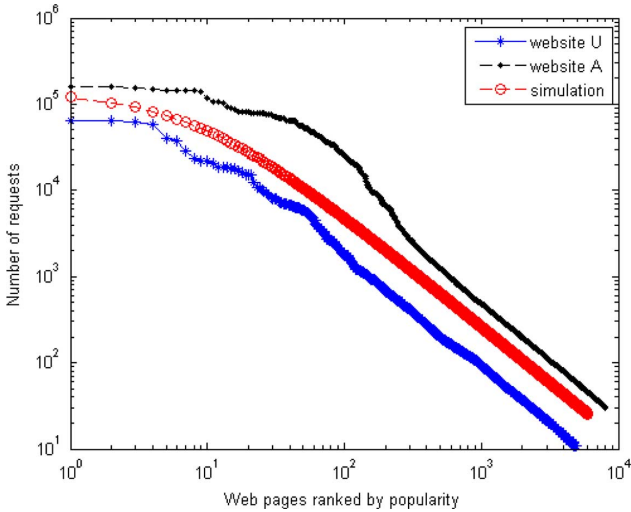


Fig. 6. Size distribution of web object for two popular Chinese web sites.

Fig. 7. Web page popularity distribution in terms of number of requests for two popular Chinese web sites. Simulation curve is with $\alpha_z = 1.2$ and $q = 8$.

lated the cumulative probability of requests against the ranking of the web pages in terms of popularity (assume the total number of web pages is 9000). The result is shown in Fig. 8. We noted that the most popular 1000 pages (out of 9000) covers more than 75% of the total request probability, and the most popular 3500 pages (out of 9000) covers 90.16% of the total request probability. As a result, we took the most popular 3500 web pages from both of the data sets for the following experiments.

We have to figure out the $\bar{N}$ and $\bar{S}$ to forge the standard web pages for web site U and A, respectively. For web site U, the most popular 3500 web pages have 164 124 web objects in total, and 136 838 for web site A. In general, α_p varies for different web sites, we first calculated the $\bar{N}$ and $\bar{S}$ according to various α_p for web site U and A respectively. The results are listed in Tables II and III (the unit kb stands for kilobyte).

We note that α_z and α_p are independent as they measure different objects. The numerical variation of α_p does not have an essential impact on the results. In order to make it consistent, we set $\alpha_p = 1.2$ in the following experiments. In this case, we take

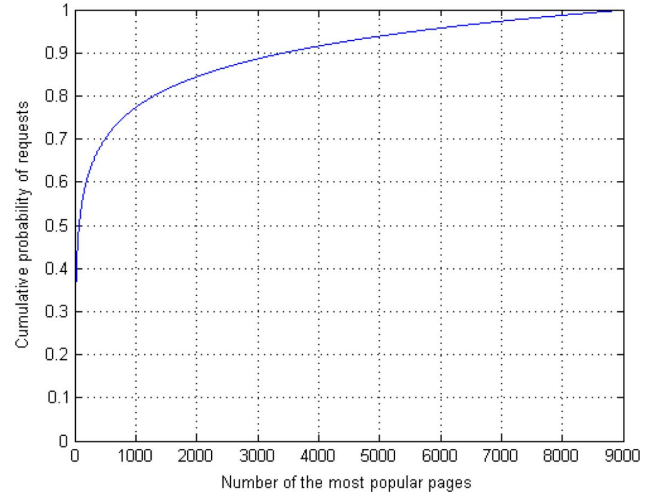


Fig. 8. Cumulative probability of request against number of web pages ranked from most popular to least (total page number is 9000).

TABLE II
VARIATION OF $\bar{N}$ AND $\bar{S}$ UNDER DIFFERENT α_p FOR WEB SITE U

α	0.6	0.8	1.0	1.2	1.4	1.6	1.8
$\bar{N}$	45.77	43.39	39.94	37.13	36.61	38.12	40.78
$\bar{S}(kb)$	52.93	54.81	58.76	66.00	76.17	87.63	98.97

TABLE III
VARIATION OF $\bar{N}$ AND $\bar{S}$ UNDER DIFFERENT α_p FOR WEB SITE A

α	0.6	0.8	1.0	1.2	1.4	1.6	1.8
$\bar{N}$	31.39	27.29	24.64	25.57	29.87	35.66	41.44
$\bar{S}(kb)$	22.21	23.33	27.02	33.93	42.74	51.59	59.49

$\bar{N} = 37.13$ and $\bar{S} = 66.00$ kb for web site U, and $\bar{N} = 25.57$, $\bar{S} = 33.93$ kb for web site A. In other words, the size of a standard page is about 2451 kb for web site U, and 868 kb for web site A.

Following the statistical results regarding the browsing length and standard page size, in order to make a browsing session enjoy an “infinite” cache environment for predicted packet padding, a client has to offer around 70 megabytes for web site U, and around 20 megabytes for web site A. This memory space requirement can be easily met by today’s personal computers and notebooks. However, it is a challenge for some wireless mobile devices.

Finally, we started to investigate the benefit that we could gain by using the proposed predicted packet padding strategy. As shown in (25), the CCA of dummy packet padding is around 2 on average, and this is the target that we try to beat. We attempted to see how the proposed strategy performed in terms of CCA. First of all, we studied the variation of the CCA against browsing length in the case of infinite cache size.

We randomly took 50 browsing sessions from each data set respectively (they belong to multiple users) and calculated the CCA for each web site. Moreover, the CCA of the dummy packet padding strategy is also calculated respectively for each web site for comparison purposes. The results are presented in Fig. 9. This experiment confirmed our theoretical analysis in the following aspects.

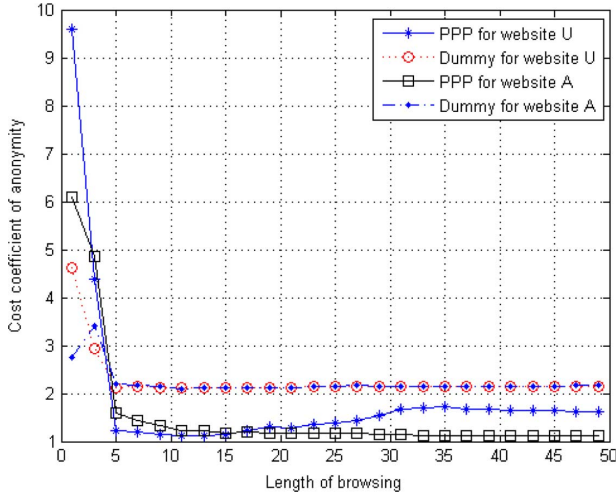


Fig. 9. Variation of cost coefficient of anonymity against length of browsing under infinite cache size condition (PPP stands for predicted packet padding, and dummy represents dummy packet padding).

- 1) The CCA of dummy packet padding converges quickly to around 2 when browsing length is greater than 5, which is consistent with our theoretical analysis.
- 2) In general, the CCA of the proposed padding strategy decreases when browsing length increases, which confirms the advantages of the proposed method. This phenomenon is clearly confirmed by the case of web site A. However, the CCA increases for web site U when the browsing length is more than 20. This is caused by the decrease of the prediction accuracy. However, it is always less than 2, which means the proposed method beats the traditional dummy packet padding strategy when the browsing length is sufficient.
- 3) The proposed predicted packet padding strategy beats the traditional dummy packet padding method when browsing length is greater than 5, in other words, in general the proposed method outperforms the dummy packet padding one as web viewers usually access more than 5 web pages for a browsing session of a web site.

We can further interpret the results of the experiments from other two perspectives: the bandwidth and the delay.

From the bandwidth aspect, the experiments indicated that the waste of bandwidth decreases when a user browses more and more web pages at a web site, which means more and more prefetched web pages are requested. In other words, the more web pages a user has browsed, the less waste of bandwidth.

The phenomenon of end to end delay is similar. At the beginning of a browsing session, the delay of the proposed method is similar to that of using dummy packet padding, around two times the delay of browsing without padding. However, the average end to end delay of the proposed method approaches the delay of browsing without packet padding when a viewer browses more and more pages, while the end to end delay of the dummy packet padding remains the same. In other words, if a viewer browses a sufficient number of web pages, the proposed method achieves half delay compared to that of dummy packet padding.

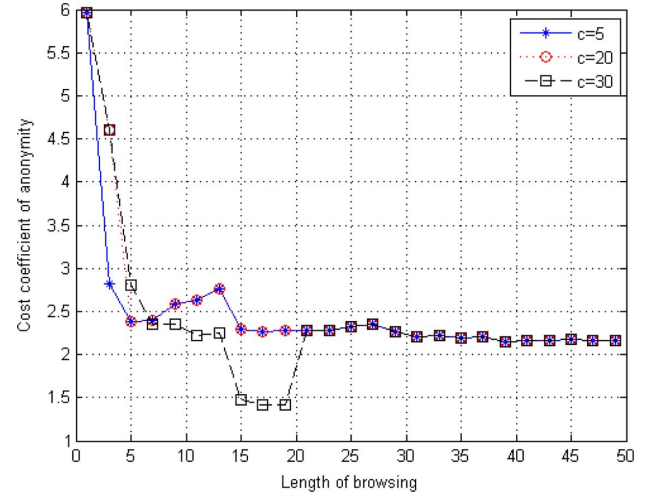


Fig. 10. Impact on cost coefficient of anonymity of proposed method against length of browsing under finite cache size conditions of relative cache size as 5, 20 and 30, respectively.

We further examined the impact on CCA of the length of browsing with different cache sizes. We randomly took 50 browsing sessions from the data set of web site A because the previous experiment indicated that it aligned well with the theoretical analysis. We calculated the CCA under three circumstances, $c = 5, 20$ and 30 . The result is shown in Fig. 10. The result is consistent with our theoretical analysis: when browsing length is greater than the relative cache size c , the CCA of the proposed method is similar to the dummy packet padding method.

VIII. FURTHER DISCUSSION

In this paper, we proposed the predicted packet padding method to make anonymous web browsing feasible in practice. Our goal was to show the effectiveness of the proposed method. However, it is hard and expensive to offer anonymous communication in reality as any vulnerable component of a system can be used by attackers to break the system anonymity. We discuss some of them in this section.

- 1) Dynamic web pages. We have investigated the effectiveness of the proposed method for static web pages, however, a large number of web pages are dynamic pages, e.g., e-business web pages. The pages are dynamic for legitimate users and attackers. It is worthwhile to study the proposed method in this case.
- 2) Attacks based on page request time intervals. In this paper, we discussed the protection for page padding at the server side. Actually, a browsers' browsing pattern, e.g., the page request time interval, is also vulnerable to attackers. For example, web viewers usually follow the request-view-request pattern, and the time interval they view pages depend on the length of the pages. Based on this kind of information, attackers can use the hidden Markov or the semi-hidden Markov model to break the anonymity.
- 3) Link padding. In this paper, we only researched the case of page padding and neglected link padding. As we know, the information on the communication channel also releases

rich information to attackers. The current link padding strategy uses dummy packets to keep the communication channel the same, namely, for any time interval, the traffic is the same (maybe intended traffic or dummy traffic). Obviously, it is very expensive to implement link padding.

Moreover, the good news is that the success of attackers depends on how much information they can obtain. In practice, it is also hard (even harder) for attackers to collect sufficient information to crack our systems. Therefore, we just need to make our systems a bit stronger than the attackers' capability, rather than implement a perfect system. We believe game theory is an excellent candidate for the research in this direction.

IX. SUMMARY AND FUTURE WORK

In this paper, we focused on reducing the delay and bandwidth waste of anonymous web browsing systems in order to make anonymous web browsing applicable for web viewers. We proposed the predicted packet padding strategy to achieve this goal. A simple mathematical model for the packet padding mechanism was established, followed by a thorough analysis and comparison between the proposed strategy and the traditional dummy packet padding method. Moreover, a metric, CCA, was defined to measure the performance of different packet padding strategies. In theory, we found that the CCA of traditional dummy packet padding is around 2, and the CCA of our predicted padding strategy approaches 1 when browsing length is sufficient, e.g., greater than 15, which means there was very limited bandwidth waste and very limited extra delay under the proposed padding strategy. Our real data set experiments confirmed our theoretical conclusions.

As for our future work, there are a few directions to explore further. First of all, we would like to extend our research to dynamic web site case as we did not involve it in the current paper. Secondly, we would like to implement our strategy in a real anonymous communication platform, such as Tor, to test our strategy in real practice. Thirdly, because more and more people use mobile devices, such as smart phones, to access the web, and these devices have much smaller cache memories than that of traditional computers, further research is needed to address problems in this direction.

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REFERENCES

- [1] M. Edman and B. Yener, "On anonymity in an electronic society: A survey of anonymous communication systems," *ACM Computing Survey*, vol. 42, no. 1, 2009.
- [2] D. Chaum, "Untraceable electronic mail, return addresses, and digital pseudonyms," *Commun. ACM*, vol. 24, no. 2, pp. 84–88, 1981.
- [3] D. M. Goldschlag, M. G. Reed, and P. F. Syverson, "Hiding routing information," in *Information Hiding*, 1996, pp. 137–150.
- [4] M. G. Reed, P. F. Syverson, and D. M. Goldschlag, "Anonymous connections and onion routing," *IEEE J. Select. Areas Commun.*, vol. 16, no. 2, pp. 482–494, Feb. 1998.
- [5] [Online]. Available: <http://www.torproject.org>
- [6] R. Dingledine, N. Mathewson, and P. F. Syverson, "Tor: The second-generation onion router," in *Proc. USENIX Security Symp.*, 2004, pp. 303–320.
- [7] M. K. Reiter and A. D. Rubin, "Crowds: Anonymity for web transactions," *ACM Trans. Inform. Syst. Security*, vol. 1, no. 1, pp. 66–92, 1998.
- [8] C. Diaz, S. Seys, J. Claessens, and B. Preneel, R. Dingledine and P. Syverson, Eds., "Towards measuring anonymity," in *Proc. Privacy Enhancing Technologies Workshop (PET 2002)*, Apr. 2002, Springer-Verlag, LNCS 2482.
- [9] A. Serjantov and G. Danezis, R. Dingledine and P. Syverson, Eds., "Towards an information theoretic metric for anonymity," in *Proc. Privacy Enhancing Technologies Workshop (PET 2002)*, LNCS 2482, Apr. 2002, Springer-Verlag.
- [10] Q. Sun, D. R. Simon, Y.-M. Wang, W. Russell, V. N. Padmanabhan, and L. Qiu, "Statistical identification of encrypted web browsing traffic," in *Proc. IEEE Symp. Security and Privacy*, 2002.
- [11] M. Liberatore and B. N. Levine, "Inferring the source of encrypted http connections," in *Proc. 13th ACM Conf. Computer Communications Security*, New York, 2006, pp. 255–263, ACM.
- [12] Y. Zhu, X. Fu, B. Gramham, R. Bettati, and W. Zhao, "Correlation-based traffic analysis attacks on anonymity networks," *IEEE Trans. Parallel Distributed Syst.*, vol. 21, no. 7, pp. 954–967, Jul. 2010.
- [13] W. Yu, X. Fu, S. Graham, D. Xuan, and W. Zhao, "DSSS-based flow marking technique for invisible traceback," in *Proc. IEEE Symp. Security Privacy*, 2007, pp. 18–32.
- [14] R. Pries, W. Yu, X. Fu, and W. Zhao, "A new replay attack against anonymous communication networks," in *Proc. ICC*, 2008, pp. 1578–1582.
- [15] W. Jia, P. Tso, X. Fu, Z. Lin, D. Xuan, and W. Yu, "Blind detection of spread spectrum flow watermarks," in *Proc. 28th Conf. Computer Commun.*, 2009.
- [16] Z. Ling, J. Luo, W. Yu, X. Fu, D. Xuan, and W. Jia, "A new cell counter based attack against Tor," in *Proc. ACM Conf. Computer Communications Security*, 2009, pp. 578–589.
- [17] A. Hintz, "Fingerprinting websites using traffic analysis," in *Proc. Workshop Privacy Enhancing Technologies*, 2002.
- [18] P. Venkatasubramanian, T. He, and L. Tong, "Anonymous networking amidst eavesdroppers," *IEEE Trans. Inform. Theory*, vol. 54, no. 6, pp. 2770–2784, Jun. 2008.
- [19] C. E. Shannon, "Communication theory of secrecy systems," *J. Bell Syst. Technol.*, vol. 28, pp. 656–715, 1949.
- [20] S. Yu, T. Thapngam, S. Wei, and W. Zhou, "Efficient web browsing with perfect anonymity using page prefetching," in *Proc. 10th Int. Conf. Algorithms and Architectures for Parallel Processing (ICA3PP 2010)*, 2010, pp. 1–12.
- [21] W. Wang, M. Motani, and V. Srinivasan, "Dependent link padding algorithms for low latency anonymity systems," in *Proc. ACM Conf. Computer and Communications Security*, 2008, pp. 323–332.
- [22] P. Venkatasubramanian and L. Tong, "A game theoretic approach to anonymous networking," *IEEE Trans. Networking*, to be published.
- [23] B. A. Huberman, P. L. T. Pirolli, J. E. Pitkow, and R. M. Lukose, "Strong regularities in world wide web surfing," *Science*, vol. 280, no. 3, 1998.
- [24] Y. Xie and S.-Z. Yu, "A large-scale hidden semi-Markov model for anomaly detection on user browsing behaviors," *IEEE/ACM Trans. Networking*, vol. 17, no. 1, pp. 54–65, Jan. 2009.
- [25] W.-G. Teng and C.-Y. Chang, "Integrating web caching and web prefetching in client-side proxies," *IEEE Trans. Parallel Distributed Syst.*, vol. 16, no. 5, pp. 444–455, May 2005.
- [26] M. Awad, L. Khan, and B. Thuraisingham, "Predicting www surfing using multiple evidence combination," *VLDB J.*, vol. 17, no. 3, pp. 401–417, 2008.
- [27] [Online]. Available: <http://www.w3.org/Protocols/rfc2616/rfc2616.html>
- [28] X. Fu, B. Graham, R. Bettati, and W. Zhao, "On countermeasures to traffic analysis attacks," in *Proc. IAW*, 2003, pp. 188–195.
- [29] C. V. Wright, F. Monrose, and G. M. Masson, "On inferring application protocol behaviors in encrypted network traffic," *J. Machine Learning Res.*, vol. 7, pp. 2745–2769, 2006.
- [30] S. E. Coull, M. P. Collins, C. V. Wright, F. Monrose, and M. K. Reiter, "On web browsing privacy in anonymized netflows," in *Proc. 16th USENIX Security Symp.*, Berkeley, CA, 2007, pp. 1–14.
- [31] C. V. Wright, S. E. Coull, and F. Monrose, "Traffic morphing: An efficient defense against statistical traffic analysis," in *Proc. NDSS*, 2009.
- [32] A. L. Montgomery, S. Li, and K. Srinivasan, "Modeling online browsing and path analysis using clickstream data," *Marketing Sci.*, vol. 23, no. 4, pp. 579–595, 2004.

- [33] L. Breslau, P. Cao, L. Fan, G. Phillips, and S. Shenker, "Web caching and zipf-like distributions: Evidence and implications," in *Proc. INFOCOM*, 1999, pp. 126–134.
- [34] Z. K. Silagadze, "Citations and the Zipf-Mandelbrot's law," *Complex Syst.*, vol. 11, pp. 487–499, 1997.
- [35] M. Hefeeda and O. Saleh, "Traffic modeling and proportional partial caching for peer-to-peer systems," *IEEE/ACM Trans. Networking*, vol. 16, no. 6, pp. 1447–1460, Jun. 2008.
- [36] A. Klemm, C. Lindemann, M. K. Vernon, and O. P. Waldhorst, "Characterizing the query behavior in peer-to-peer file sharing systems," in *Proc. 4th ACM SIGCOMM Conf. Internet Measurement*, 2004, pp. 55–67.
- [37] M. E. Crovella and A. Bestavros, "Self-similarity in world wide web traffic: Evidence and possible causes," *IEEE/ACM Trans. Networking*, vol. 5, no. 6, pp. 835–846, Jun. 1997.
- [38] J. Czyzowicz, E. Kranakis, D. Krizanc, A. Pelc, and M. V. Martin, "Enhancing hyperlink structure for improving web performance," *J. Web Eng.*, vol. 1, no. 2, pp. 93–127, 2003.
- [39] W. J. Reed and M. Jorgensen, "The double pareto-lognormal distribution—A new parametric model for size distributions," *Commun. in Statistics—Theory and Methods*, vol. 33, no. 8, pp. 1733–1753, 2003.
- [40] S. Burklen, P. J. Marron, S. Fritsch, and K. Rothermel, "User centric walk: An integrated approach for modeling the browsing behavior of users on the web," in *ANSS'05: Proc. 38th Annu. Symp. Simulation*, 2005, pp. 149–159, IEEE Computer Society.
- [41] M. E. J. Newman, "Power laws, Pareto distributions and Zipf's law," *Contemporary Phys.*, vol. 46, pp. 323–351, Dec. 2005.



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